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A simheuristic approach using the NSGA-II to solve a bi-objective stochastic flexible job shop problem

Camilo Andrés Rodríguez-Espinosa, Eliana María González-Neira and Gabriel Mauricio Zambrano-Rey

Departamento de Ingeniería Industrial, Facultad de Ingeniería, Pontificia Universidad Javeriana, Bogotá, Colombia

ABSTRACT

This paper addresses a bi-objective problem in flexible job shop scheduling (FJSS) with stochastic processing times. Following the Just-In-Time philosophy, the first objective is to minimise deterministic Earliness+Tardiness, and the second objective is to minimise the Earliness+Tardiness Risk. The second objective function seeks to obtain robust solutions under uncertain environments. The proposed approach is a simheuristic that hybridises the non-dominated sorting genetic algorithm (NSGA-II) with Monte Carlo simulation to obtain the Pareto frontier of both objectives. The computational results demonstrate the effectiveness of the proposed algorithm under different variability environments.

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Stochastic Flexible Job Shop Scheduling; Earliness +tardiness; Simulation-optimisation; Robustness; Non-dominated sorting genetic algorithm (NSGA-II)

1. Introduction

Production scheduling is a decision-making process that involves the allocation of resources to tasks over given time periods to achieve one or more specific goals. Particularly, the job shop scheduling (JSS) problem is an environment in which each job has a particular route of operations that must be followed and each of those can be processed only by a single machine. The generalisation of this environment is the flexible job shop scheduling (FJSS), in which each operation can be processed in one or more machines (Pinedo, 2016). Due to its combinatorial nature, the JSS and, consequently, the FJSS are classified as NP-hard (Demir & İşleyen, 2013; Y. Yang et al., 2020). The FJSS problem exists in many important industries, such as chemical material processing (Kacem et al., 2002), aeronautical (J. Zhang & Yang, 2016), semiconductor manufacturing (Tamssaouet et al., 2022), and steelmaking (J.-Q. Li et al., 2020), among others. Following the just in time (JIT) philosophy, which traditionally implies not finishing jobs too early, to prevent increments in finished product inventory costs; and not delivering jobs too late, to avoid incurring penalties with customers, the minimisation of the sum of Earliness and tardiness is considered as the main objective function of the study (Gonzalez-Neira et al., 2021; Horng et al., 2012).

Seeking to have a representation that more closely resembles reality, the performance of a schedule can be influenced by stochastic factors, such as machine breakdowns, the release of jobs or processing times of operations (Horng et al., 2012; Pinedo, 2016; Shen et al., 2017). In this case, the stochastic version of the FJSS is studied considering stochastic processing times because, as stated by Pinedo (2016), other sources of uncertainty, such as machine breakdowns, can be modelled as an internal part of the processing times. There are three main approaches for scheduling under uncertainties: reactive, predictive, and predictive-reactive (Aytug et al., 2005; Z. Li & Ierapetritou, 2008). Reactive scheduling sequences jobs with real-time dispatching algorithms that create partial schedules based on limited and specific information. Predictive methods consist in creating a schedule that minimises the effects of disruptions. In this category, five approaches have been reported in the literature: stochastic, robust, fuzzy, sensitivity analysis, and parametric programming (Z. Li & Ierapetritou, 2008). Predictive-reactive techniques combine the predictive and reactive approaches, but this process must deal with the right timing to start the rescheduling action, and also with the action that should be taken. For this work, the predictive approach is chosen because its results are the basis for planning support activities, which allows the estimation of completion times and costs.

Considering the stochastic environment, robust scheduling is considered to minimise the effects of the uncertainties in the predictive schedule. For this purpose, the robust measure Earliness+Tardiness Risk is used to ensure that the predictive schedule does not differ excessively from the realised schedule, maintaining a high-level performance of the objective. In that way, the obtained schedule uses the limited resources efficiently and allows sufficient flexibility for changes (Al-Hinai & ElMekkawy, 2011).

CONTACT Camilo Andrés Rodríguez-Espinosa camiloa.rodriguez@javeriana.edu.co Departamento de Ingeniería Industrial, Facultad de Ingeniería, Pontificia Universidad Javeriana, Bogotá, Colombia

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One way to address the presence of uncertainty in complex optimisation problems is through simheuristics, which are the combination of simulation techniques and metaheuristic algorithms (Juan et al., 2015). Simheuristics have been successfully applied for solving vehicle routing (González-Martín et al., 2016; Juan et al., 2011), and production scheduling (González-Neira et al., 2019; Hatami et al., 2018; Juan et al., 2014), among others. Simheuristics are extremely flexible, allowing the integration of different metaheuristic algorithms and simulation techniques (Juan et al., 2015). According to Chica et al., (2017) the paradigms of simulation that can be used to build a simheuristic are Monte Carlo, discrete event, system dynamics, and agent-based modelling. Of these, the Monte Carlo simulation requires the least computational consumption. For that reason, Monte Carlo paradigm is selected for this study.

Considering that the problem to be solved has two objectives, a multi-objective optimisation approach has to be considered. Although these types of problems address real-life scenarios better, they include an additional difficulty in terms of handling their objectives, which have to be analysed in terms of the trade-off between them, called the Pareto optimal set (Türkyılmaz et al., 2020; Von Lücken et al., 2014). Different techniques have been implemented to solve this problem, such as the weighted sum method, hierarchical optimisation method, lexicographic ordering, successive Pareto optimisation, methods based on evolutionary algorithms, and methods based on swarm intelligence, among others (Türkyılmaz et al., 2020). For this case, the non-dominated sorting genetic algorithm II (NSGA-II) is proposed to obtain the Pareto frontier of the objective functions using Monte Carlo simulation to evaluate the performance of the robust measure among different random scenarios (Caldeira & Gnanavelbabu, 2021). NSGA-II is characterised by its low computational requirements, elitist approach and parameter-less sharing approach to improve solutions in different applications, obtaining better results than the Pareto-archived evolutionary strategy (PAES) and the strength Pareto evolutionary algorithm (SPEA) in terms of the spread of solutions and convergence near the true Pareto-optimal front (Deb et al., 2002) and better results than the non-dominated ranking genetic algorithm (NRGA) in terms of the spacing, convergence of the Pareto front and quantity of obtained solutions in a stochastic FJSS problem (Ahmadi et al., 2016). The NSGA-II simultaneously optimises each objective without being dominated by any other solution (Yusoff et al., 2011). The NSGA-II has great performance with a small number of objectives but as this number increases the performance of this algorithm gets worse (Von Lücken et al., 2014). Some areas where the NSGA-II has been implemented are: finance-based scheduling (El-Abbasy et al., 2020), construction (Bagheri-Esfeh et al., 2020), logistic distribution network (Wang et al., 2021), and so forth.

The rest of the paper is organised as follows. Section 2 is dedicated to reviewing studies related to stochastic FJSS (SFJSS) and/or robust FJSS. In Section 3, the problem formulation is presented. In Section 4, the proposed simheuristic is explained. The results of the simheuristic performance are reported in Section 5, and then the conclusions and future work are stated in Section 6.

2. Literature review

Since the aim of this paper is to solve a stochastic and robust FJSS through a simheuristic, this section is divided into three parts. The first subsection reports some literature related to stochastic and/or robust FJSS problems. The second part presents some works related to the application of simheuristics in optimisation problems other than scheduling ones. The third subsection states the elements on which this study focuses on and its contribution.

2.1. Literature in stochastic and/or robust FJSS

A search was executed on the SCOPUS and ISI Web of Science databases, using the following terms: ("Stochastic" OR "Robust") AND ("Flexible Job Shop Scheduling"). Around 50 papers were found, and after filtering out those related to job shops and merely reactive approaches, only 20 papers, dealing with the flexible job shop problem, are reported hereafter. The following analysis is presented considering the source of uncertainty, the objective functions, the method used to build the schedule, the robust measure and the insertion of simulation to account for the stochastic behaviour. Based on our findings, the literature on stochastic and/or robust FJSS can be presented in two major groups. In the first group of papers (subsection 2.1.1), it can be seen that the objective function does not include any robust or stability performance metrics. On the contrary, the second group (subsection 2.1.2) includes those methods that aim to minimise not only classic production-oriented metrics (e.g., makespan, flowtime or due-date related functions) but also robustness and stability functions in order to produce schedules that consider the impact of uncertain events. Both groups are analysed to determine if any of the paradigms of simulation has been integrated into the solution approach. Those papers that have integrated any type of simulation with a metaheuristic can be classified as research that has implemented a simheuristic approach, and this represents 45% of the stochastic and/or robust literature consulted.

2.1.1. Objective functions without robustness/stability measures

Within this category, the reported papers are also organised depending on the use of simulation to account for uncertainty (see Table 1). For instance, Ayyoubzadeh et al. (2021) address the SFJSS under two kinds of uncertainty. The first kind, tries to reproduce the impact of resource unavailability, the lack of skilled workers and inadequate estimation of operations, which can be modelled using pre-known probability distributions affecting the start and processing times of the jobs. Conversely, certain parameters, such as machine breakdowns, order cancellation or modification, as well as new order arrivals may not follow well-known probability distributions; hence, a reactive scheduling strategy is proposed for this type of disruption. The authors proposed a non-dominated sorting genetic algorithm (NSGA-II) to deal with two conflicting objective functions: minimising the tax cost on surplus energy consumption and minimising the total cost of jobs lateness based on soft time-windows. The chromosome structure is divided into two sub-chromosomes: the first sub-chromosome, focuses on the stochastic start times, modelled using a uniform distribution. The second sub-chromosome, accommodates minor disruptions based on the original schedule set by the first sub-chromosome.

Similarly, X. Yang et al. (2016) proposed a NSGA-II to tackle stochastic processing times, aiming to minimise the completion time and total energy consumption. The NSGA-II first finds a basic scenario, and then, using parameter sensitivity analysis, the change of average values of the objective function is observed based on disturbed processing times, subject to a uniform distribution, with different degrees of severity. The analysis determines the level of machine redundancy to absorb such perturbations and its consequences in performance loss of the scheduling scheme.

Likewise, Jin et al. (2021) proposed a population-based metaheuristic approach to deal with the SFJSS under stochastic processing times modelled using triangular neutrosophic numbers. Neutrosophic sets, unlike fuzzy sets, quantify the truth, indeterminacy and falsity memberships, instead of using only one membership function. Then, the neutrosophic numbers were used to develop an improved teaching-learning-based optimisation (TLBO) algorithm, using genetic operators, such as crossover. The performance of the proposed approach was studied under small, medium and large uncertainty scenarios, to configure the neutrosophic processing times.

Hybrid simulation-optimisation techniques have been used for decades, but flourished in the last 20 years, mainly because of the increase in computational power. Particularly, in simulation-based optimisation, the optimisation algorithm controls the execution by calling the simulation model whenever needed (Juan et al., 2015). This is the case of simheuristic approaches, also known as optimisation via simulation. For example, Gocken et al. (2019) proposed an optimisation via simulation approach for the SFJSS. The optimisation part is based on a genetic algorithm (GA), which calls upon a discrete-event simulation (DES) model of the system, coded in SIMIO 6.1. Process plans and machine assignments are explored by the GA and sent as decision variables to the simulation model, which returns the objective function value. The arrival rate of parts is exponentially distributed. Also using DES, Gupta and Jain (2022) proposed a simulation-based optimisation approach for simultaneous preventive maintenance, and random machine breakdown environment with sequence-dependent setup time (SDST). For the optimisation part, the desirability function approach is used to optimise four different performance measures, i.e., mean flowtime, makespan, mean tardiness, and total setups. In turn, the simulation model (implemented in PROMODEL) is configured to represent the stochastic behaviour of the system in terms of job arrivals following a Poisson distribution, uniformly distributed processing times, and machine availability described by two parameters, the mean time between failures (MTBF) and the mean time to repair (MTTR), which are exponentially distributed. As mentioned by Gocken et al. (2019), using DES enables the representation of complex

Table 1. Works that do not consider robustness/stability measures.

Authors	Source of Uncertainty	Objective Function	Method	Use of simulation?
Ayyoubzadeh et al. (2021)	Start time Processing time Machine breakdowns Order cancellation Order modification Order arrivals	*Min* Tax cost on surplus energy consumption *Min* Total cost of jobs lateness	NSGA-II	No, random times inserted in chromosome
X. Yang et al. (2016)	Processing times	*Min* Completion time *Min* Total energy consumption	NSGA-II	No, parameter sensitivity analysis based on uncertainty levels
Jin et al. (2021)	Processing times	*Min* Makespan (C_{max})	TLBO	No, Single-valued neutrosophic sets
Gocken et al. (2019)	Part arrivals	*Min* Total average flowtime of each part	GA	Yes, discrete-event simulation
Gupta and Jain (2022)	Machine breakdowns Processing times	*Min* Mean flow time *Min* Makespan (C_{max}) *Min* Mean tardiness *Min* Total setups	Desirability function approach	Yes, discrete-event simulation

environments in which it is hardly to obtain tractable mathematical formulations. However, coupling simulation models to optimisation requires much more time to evaluate each scenario, risking high computational costs.

Another way to simulate sample scenarios and obtain estimates of objective functions is by coupling Monte Carlo simulation to an optimisation method. Monte Carlo simulation is used to determine which scenario, among the best-found deterministic ones, shows the best objective function considering stochastic parameters (Juan et al., 2015). As concerns the optimisation method, heuristic and metaheuristic approaches are usually implemented. For example, Dosdoğru et al. (2015) studied the impact of routing flexibility on the expected makespan, within a FJSS, integrating a genetic algorithm with Monte Carlo simulation. Processing times were randomly generated using a triangular distribution, and the number of operations was uniformly distributed. Focusing on the same objective function, Lin et al. (2022) implemented a learning-based grey wolf optimiser to schedule jobs in a semiconductor manufacturing environment. The processing times following a gamma distribution are also generated with Monte Carlo simulation. On the same lines, Caldeira and Gnanavelbabu (2021) modelled stochastic processing times with a log-normal distribution, to reproduce non-negative values. The authors applied the Jaya algorithm due to the absence of algorithm-specific parameters that require to be tuned. Thus, experiments can focus on the effect of stochastic processing times without the interference of the optimisation algorithm's parameters.

As mentioned before, another source of uncertainty that has attracted interest is machine availability. Rahmati et al. (2018) considered two maintenance scenarios in terms of corrective maintenance (CM), caused by machine disruptions between inspection intervals, and preventive maintenance (PM). The event time activity is triggered by a stochastic variable that denotes shock or the machine degradation level and follows an exponential distribution. The stochastic time between two shocks is also exponentially distributed. Moreover, the duration of PM activities, the recovery level through PM and the duration of CM all follow log-normal distributions, with different parameters. To achieve this, the harmony search optimisation algorithm is coupled with an optimisation module (not explicitly stated as Monte Carlo simulation but it can be assumed) to find robust schedules aiming to minimise the expected makespan (C_{max}). This problem shows the power of simulation-based optimisation (i.e., simheuristics) to study various sources of uncertainty within a complex manufacturing setting.

2.1.2. Objective functions considering robustness/stability measures

The most significant and widespread objectives address minimising the completion and flow times, or due-date related functions (see Table 2). Hence, schedules generated to meet such purposes tend to be tight and lack responsiveness to uncertainties. Therefore, the degradation and stability of the objective in the presence of disruptions, can be integrated as measures of a schedule's robustness, among other similar metrics (Sajadi et al., 2019). Thus, a definition of degradation or regret can be related to the deviation between the expected value of a performance measure (e.g., makespan), under a specific scenario, and its corresponding optimal or pre-established value. Inspired by this robustness measure, Hao et al. (2014) proposed a Markov network-based estimation of distribution algorithm to solve the SFJSS under uncertain processing times. The processing times follow a uniform distribution within defined lower and upper bounds. So, processing time uncertainty can be constructed through the concept of a scenario that corresponds to the assignment of a reasonable processing time to each operation on each machine. Then, a finite set of scenarios is sampled from a set of random processing times generated with the Cartesian product. For each scenario, the expected regret value is estimated, and several scenarios are carried out to minimise the maximum expected regret. Alternatively, a variety of processing times scenarios can be set in advance to calculate a set of scenarios. Bearing this in mind, Shen et al. (2017) developed a modified multi-objective evolutionary algorithm based on decomposition (m-MOEA/D) with two scenario-based robustness measures based on the relative variation between uncertain makespan and workload, and their corresponding deterministic values.

Scenario-based approaches calculate the deviations between the efficiency performances obtained from various scenarios, and the performance produced in the initial deterministic scenario. Various works have applied this strategy by proposing two-stage approaches. Taking this into account, Nouiri et al. (2017) proposed a two-stage particle swarm optimisation (PSO) to solve the problem considering schedule stability. In the first stage, the PSO focuses on minimising only the makespan, assuming deterministic parameters, meaning no disruptions. Then, the second stage takes the final population of the PSO as the initial one, to improve a bi-objective function composed of a weighted sum of the makespan and a stability measure. This latter is similar to the regret value mentioned before, because its calculated as the difference between the completion times found in the first stage and those found in the second stage. Machine breakdowns are inserted into the second

Table 2. Works with objective functions considering robustness/stability measures.

Authors	Source of Uncertainty	Main Objective Function	Method	Robustness measures	Use of simulation?
Shen et al. (2017)	Processing times	*Min* Makespan (C_{max}) *Min* − −*Max* Workload	m – MOEA/D	Relative variety of Makespan and workload	No, random processing times inserted into the optimisation method
Nouiri et al. (2017)	Machine breakdowns	*Min* Makespan (C_{max})	2S – PSO	Stability = *Min* Difference of completion times	No, scenario – based approach on the second stage
Al-Hinai and ElMekkawy (2011)	Machine breakdowns	*Min* Makespan (C_{max})	2S – HGA	Stability	No, scenario – based approach on the second stage
He and Sun (2013)	Machine breakdowns	*Min* Makespan (C_{max})	GA plus Right-Shift Route Changing	Robustness Stability	No, scenario – based approach on the second stage
Thi et al. (2022)	Machine breakdowns	*Min* Makespan (C_{max})	GA Three re – scheduling policies	Robustness Stability	No, scenario – based approach on the second stage
Buddala and Mahapatra (2019)	Machine breakdowns	*Min* Makespan (C_{max})	2S – TLBO	Robustness Stability	
Ahmadi et al. (2016)	Machine breakdowns	*Min* Makespan (C_{max})	NSGA – II NRGA	Stability = *Min* Difference of completion times	Yes, incorporated simulation module
Sajadi et al. (2019)	Machine breakdowns	*Min* Realised Makespan (C_{max})	GA – NSGA–II NRGA	Stability = *Min* Difference of completion times	Yes, incorporated simulation module
Xiong et al. (2013)	Machine breakdowns	*Min* Makespan (C_{max})	NSGA – II	SM1 (Workload, float times) SM2 (distance between central points)	Yes, Monte Carlo simulation
Mokhtari and Dadgar (2015)	Processing times Due-dates	*Min* Tardy jobs	MILP Simulated Annealing (SA)	Machine availability	Yes, Monte Carlo simulation
Gnanavelbabu et al. (2021)	Processing times Worker flexibility	*Min* Makespan (C_{max})	MBSA	*Min* Std C_{max}	Yes, Monte Carlo simulation
Gomez et al. (2021)	Processing times		TS	*Max* Service level	Yes, Monte Carlo simulation
Y. Yang et al. (2020)	Machine breakdowns	*Min* Makespan (C_{max})	NSGA – II	RMc based on the ELM	Yes, simulation-based method
G. Zhang et al. (2022)	Machine breakdowns	*Min* Makespan (C_{max})	ICA	RMn based on CNN	

stage to evaluate the effect of such disruptions on the predictive schedule performance. It is assumed that information about the uncertainty of machine breakdowns is known in advance. Several scenarios are created based on which machine breaks down, when it occurs and how long the machine is unavailable. The first value is found using an empirical relation related to the machines' workload, and the second and third values are uniformly distributed with different parameters. Instead of using PSO, Al-Hinai and ElMekkawy (2011) implemented a two-stage hybrid genetic algorithm (HGA) to deal with random machine breakdowns. The computational cost of scenario-based approaches depends closely on the number of scenarios, and their efficiency is related to the type scenario variety and representation of the actual behaviour of the system (e.g., long and short duration, early or late disruptions, among others). For instance, in the latter, 192000 test problems were evaluated in the second-stage.

Following the same two-stage approach, He and Sun (2013) proposed an idle time insertion strategy in the pre-schedule (first stage) and a route-changing strategy combined with a right-shift policy and route-changing strategies to keep the rescheduling (second stage) as stable and robust as possible in the presence of machine disruptions. Pre-scheduling is achieved with a genetic algorithm aiming to minimise the makespan. Re-scheduling is done to minimise the weighted sum of the relative robustness and stability measures. The relative robustness measure is the deviation percentage between the predictive and the realised schedule (after the breakdown), and the stability measure is the normalised difference between the predicted and realised completion times, only for those operations affected by the breakdown. A follow-up work by D. Sun et al. (2014) included a game theory approach to find the Nash equilibrium between the robustness and stability measures. Likewise, Thi et al. (2022) explored three different adjustments in the re-scheduling stage to reduce the makespan, and ensure robustness and stability as defined by the aforementioned works. The pre-scheduled sequence with an uncertain repair time is modified by combined route-changing and right-shifting; operation insertion on unaffected machines, and the generation of a new schedule that adapts to the breakdown at the lowest cost. Buddala and Mahapatra (2019) also followed the aforementioned strategy but using a two-stage teaching-learning-based optimisation (2S-TLBO) and a rescheduling technique called modified affected operations rescheduling (mAOR) was used to generate robust schedules in the second stage.

Another way to achieve two-stage strategies is by implementing a training predictive model in the first stage and a robustness prediction scheduling in the second one, as reported by Y. Yang et al. (2020) and G. Zhang et al. (2022). The former aims to minimise the makespan and a proposed robustness measure called the RMc. The RMc was developed using an extensive learning machine (ELM) in two phases: the first phase collects training samples of schedules comprising workloads and float times, and the second phase predicts the value of the RMc. In both phases, a NSGA-II is in charge of producing the schedules. The latter follows the same strategy, although in the first stage, the training is executed by a convolutional neural network (CNN). Instead of an evolutionary approach, an improved imperialist competition algorithm (ICA) is proposed to generate training schedules. Another robustness measure was proposed, called RMn, based on machine breakdown, workload and the float time of the operation.

The aforementioned two-stage approaches are an alternative to simulation in terms of random value generation for stochastic parameters. However, Monte Carlo and discrete-event simulation are resourceful tools to generate stochasticity and therefore find robust and stable schedules according to known behaviours. Ahmadi et al. (2016) addressed the stable scheduling of a multi-objective FJSS with random machine breakdowns. Two evolutionary algorithms, NSGA-II and a non-dominated ranking genetic algorithm (NRGA), are applied to combine the improvement of the makespan and stability simultaneously. A simulation module, similar to Monte Carlo, is incorporated into the optimisation methods, to reproduce machine disruptions following exponential distributions for breakdown occurrences and repair times. As well as the previous case, Sajadi et al. (2019) also applied NSGA-II and NRGA but in the second stage, and a genetic algorithm was implemented in the first stage to focus on makespan minimisation.

Surrogate measures (SM) are designed to approximate the robustness based on some characteristic parameters, such as the float time and free slacks (Shen et al., 2017). For instance, the average float times, i.e., total slack, are computed as the difference between the earliest start time and the latest start time of an activity. Another slack-based SM is the free slack, defined as the amount of time that an activity can be delayed without delaying the start of the immediately following activity. Xiong et al. (2013) argued that slack-based measures for the robustness of a schedule focus on the structure of the schedule but neglect the uncertainty. Therefore, known information about machines can be utilised as a baseline to construct schedules, such as the probability of breakdown occurrence and the location of the breakdown for each machine. Based on those data, the authors proposed two surrogate measures: (1) the first uses the machine workload and float times, considering that the float time on a machine with a heavier workload is more important than the one with a lighter workload; and (2) the second uses the average central points between the float time and breakdown periods. A NSGA-II was implemented to study five different robustness measures, finding that the second SM performs better for both small and relatively large cases, and the first one only on small cases.

Although machine breakdowns can be studied as the main source of uncertainty, processing times give us the possibility to capture machine deterioration, setup times, and worker efficiency, among other stochastic parameters. Focused on the effect of uncertain processing times and due dates, Mokhtari and Dadgar (2015) reported a mixed integer linear programming (MILP) model with the objective of minimising the number of tardy jobs. Also, a simulation-optimisation framework based on a simulated annealing (SA) optimiser embedded with local search strategies call up a Monte Carlo (MC) simulator in such a way that the minimum value of total system availability is satisfied. In addition to processing times, Gnanavelbabu et al. (2021) presented a simheuristic approach incorporating worker flexibility to minimise the expected makespan and standard deviation of the makespan. The optimisation part is realised with a multi-objective modified backtracking search algorithm (MBSA) framework that integrates the Monte Carlo simulation. To analyse the stochastic variant, four variation levels and four probability distributions of the processing times were studied. It is interesting to note that the log-normal distribution resulted in better quality solutions, while the gamma distribution gave the worst results. The Weibull and uniform distributions exhibited competitive performances compared to a log-normal distribution.

Rather than estimating the classical makespan, Gomez et al. (2021) proposed to maximise the probability of the makespan being smaller than a predefined value, called the makespan service level. This robustness measure seeks to compensate for the uncertainty of processing times. The service level of a given schedule, constructed by a tabu search (TS) procedure, is evaluated using a Monte Carlo sampling procedure. Each operation with a stochastic processing time is sampled, a fixed number of times, according to a beta distribution to generate a set of scenarios.

2.2. Simheuristics applications in other optimisation problems

Simulation-optimisation and simulation-based optimisation offer decision-makers the possibility of dealing with complexity and uncertainty. In particular, the use of heuristic and metaheuristic approaches, for the

optimisation part, has gained a lot of interest because most real size problems are NP-hard, difficult to model and require astounding computing resources. Moreover, the lack of precise information and the variability of problem parameters make simulation models the perfect fit to heuristics/metaheuristics in order to understand and cope with the dynamics of a system. Because of those benefits, simheuristic applications are found in various areas, such as transportation, logistics, healthcare production, project management, finance and social systems (Chica et al., 2017).

For instance, Quintero-Araujo et al. (2021) tackled the capacitated location routing problem (CLRP), which integrates a facility location problem with a multi-depot vehicle routing problem, aiming to minimise the facility location and routing costs. The authors proposed a biased-randomised iterated local search metaheuristic combined with Monte Carlo to simulate stochastic demands. Also focused on location decisions, Bayliss and Panadero (2023) reported a learnheuristic algorithm that solves the temporary medical facility location and population assignment problem using a queueing simulation model. In cases related to healthcare quality of service metrics such as people's travel times, waiting and completion times are analysed besides costs. Indeed, the coronavirus pandemic came with complex problems for health care systems. Another application of simheuristics was described by Martínez-Reyes et al. (2021) to deal with the distribution of personal protective equipment (PPE) to health centres, clinics, hospitals, and intensive care units (ICUs), considering uncertain demands, distribution cost and sustainability of the health system. The proposed approach consisted of an iterated local search (ILS) algorithm combined with Monte Carlo simulation (MCS).

Freight transportation is a major problem for companies because of the costs, time and environmental issues. The inventory routing problem (IRP) combines inventory management and delivery route-planning decisions. Due to the complexity of this problem, several authors have used metaheuristic approaches to solve the deterministic case. However, replenishment and delivery depend on stochastic demands. To account for uncertainty, Gruler et al. (2020) proposed a simheuristic approach that hybridises Monte Carlo simulation within a variable neighbourhood search (VNS) algorithm to solve the multi-period IRP with stochastic customer demands. Usually, IRP problems aim at minimising inventory and routing costs respecting transportation and warehouse capacities. However, time constraints can also be relevant, particularly for battery supported vehicles. Keenan et al. (2021) propose a simheuristic algorithm that integrates Monte Carlo simulation into a strategic oscillation metaheuristic, to deal with the time capacitated arc routing problem (TCARP) with stochastic demands. Similarly, Reyes-Rubiano et al. (2019) also dealt with the routing problem of electric vehicles, with the goal of minimising the expected time-based cost required to complete the freight distribution plan. A simheuristic algorithm that combines Monte Carlo simulation with a multi-start metaheuristic was used to analyse the effect of random travel times and the use of energy safety stocks on the routing plans.

In portfolio optimisation, the objective is to minimise risk for an expected portfolio return by allocating weights to the included assets. As the number of investable assets and the number of periods to consider grows, the problem becomes NP-hard. Therefore, simheuristic approaches using Monte Carlo simulation to deal with stochastic returns modelled as random variables, coupled with genetic algorithms (Nieto et al., 2022), or variable neighbourhood search (Kizys et al., 2022) have been reported.

2.3. Focus of the study and contribution

Considering the literature reviewed, this study focuses on:

- The development of a simheuristic that integrates the NSGA-II with Monte Carlo simulation which has to deal with stochastic processing times. NSGA-II has obtained efficient results in previous studies, and Monte Carlo simulation allows the use of any probability distribution to model the stochastic parameters.
- Makespan minimisation predominates as the main objective of some authors, while the objective functions related to delivery dates have a smaller share. In this study, the minimisation of earliness makes it possible to reduce inventory costs and avoid possible product losses due to spoilage, while the minimisation of tardiness guarantees a high level of service for customers. Only one paper reported in this review considered a JIT-oriented objective.
- Processing times are considered stochastic in order to model the influence of exogenous factors on task performance. The most commonly used distributions in the literature for modelling processing times are normal, exponential and uniform distributions. Specifically, for this research, exponential and log-normal distributions were considered, because they are non-negative distributions and do not have an upper bound, allowing comparisons of the results obtained with both distributions in the same scenario. On the one hand, Trietsch et al. (2012) have demonstrated statistically that the log-normal distribution is suitable and plausible to model activity times. Moreover, this distribution has been used to model processing times in other scheduling

problems, such as byBaker and Altheimer (2012); Hatami et al. (2018); Lemos and Ronconi (2015) and Castaneda et al. (2022). On the other hand, Soroush (2010) indicates that, exponential distribution is interesting, because it has memory-less property, leading to useful analysis of stochastic scheduling problems. Exponential distribution has also been analysed in the studies by Ahmadizar et al. (2010); Jang (2002) and Azadeh et al. (2012).

- Some of the reviewed studies include a robustness measure that quantifies the effect of the uncertain parameters in the problem simultaneously with the main objective function, turning the problem into a multi-objective one. For this case, the selected robustness measure analyses, the expected increase between the predictive Earliness +Tardiness and the realised Earliness +Tardiness (known as Earliness+Tardiness Risk), which denotes the potential deterioration of the predictive Earliness+Tardiness and, consequently, the potential increase of the related costs (Wu et al., 2020; Xiao et al., 2017). None of the aforementioned studies used our proposed metrics.

Therefore, the contribution of this paper is the study of a multi-objective JIT-oriented stochastic and robust flexible job shop scheduling problem, through a population-based metaheuristic hybridised with Monte Carlo simulation. The majority of previous works on FJSS problems under uncertainties have implemented single-solution-based metaheuristics, and those few that used population-based algorithms have minimised the makespan. In fact, to the best of our knowledge, this is the first work that proposes a multi-objective simheuristic to solve a stochastic and robust FJSS with JIT-oriented measures.

3. Problem formulation

In this section, a mixed-Integer linear programming (MILP) model for the deterministic FJSS is presented, followed by the robust measure for the SFJSS and the definition of the stochastic processing times used in this study.

3.1. Flexible job shop scheduling

The model is partially based on the formulation reported by Özgüven et al. (2010). The main differences are as follows: the objective function, the additional constraints to calculate earliness and tardiness for each job, and some changes in the relation of binary variables. Another important aspect is that this model uses auxiliary variables to reduce drastically the computational costs achieving, in some cases, optimal or feasible solutions in reasonable computational times.

3.1.1. Sets

I: Set of jobs.

J_i: Set of operations of job i.

O: Set of doublets (i,j) that correspond to the j-th operation in the set J_i of job i.

M: Set of machines.

M_{ij}: Set of machines that can process the operation (i,j).

3.1.2. Parameters

PT_{ijm}: Processing time of operation (i,j) on machine m. $(i,j) \in O, \in M_{ij}$.

SW_i: Lower limit of the due window interval of job i. $i \in I$.

EW_i: Upper limit of the due window interval of job i. $i \in I$.

L: A very large number.

3.1.3. Decision variables

X_{ijm}: Binary variable that takes the value of 1 if the operation (i,j) is performed on machine m, 0 otherwise. $(i,j) \in O$, $m \in M_{ij}$.

Y_{hkijm}: Binary variable that takes the value of 1 if operation (i,j) is performed immediately after operation (h,k) on machine m, 0 otherwise.

$(i,j) \in O$, $(h,k) \in O \cup \{(0,0)\}$, $m \in M_{ij} \cap M_{hk}$.

ST_{ij}: Starting time of the operation (i,j). $(i,j) \in O$.

CT_{ij}: Completion time of the operation (i,j). $(i,j) \in O \cup \{(0,0)\}$.

T_i: Tardiness of job i. $i \in I$.

E_i: Earliness of job i. $i \in I$.

3.1.4. Constrains

Constraint set 1 limits that the operation (i,j) has to be performed in only one machine:

$$\sum_{m \in M_{ij}} X_{ijm} = 1 \qquad \forall (i,j) \in O \qquad (1)$$

Constraint set 2 blinds that the Tardiness has to be higher than the difference between the dispatching time and the upper limit of the due window:

$$T_i \geq CT_{|J_i|} - EW_i \qquad \forall i \in I \qquad (2)$$

Constraint set 3 ensures that Earliness has to be higher than the difference between the lower limit of the due window and the dispatching time:

$$E_i \geq SW_i - CT_{|J_i|} \qquad \forall i \in I \qquad (3)$$

Constraint set 4 indicates that the completion time of the operation (i,j) has to be equal to the starting time of the operation (i,j) plus the processing time in the selected machine:

$$CT_{ij} = ST_{ij} + \sum_{m \in M_{ij}} \left(X_{ijm} \times PT_{ijm}\right) \quad \forall (i,j) \in O \quad (4)$$

Constraint set 5 establishes that the starting time of the operation (i,j) has to be higher or equal than the completion time of the previous operation of that job:

$$ST_{ij} \geq CT_{i(j-1)} \qquad \forall (i,j) \in O, j \geq 2 \quad (5)$$

Constraint set 6 ensures that the starting time of the operation (i,j) has to be higher or equal than the completion time of the operation (h,k) if this operation precedes operation (i,j) on the same machine. Sometimes the predecessor can be the operation $(0,0)$ indicating that operation (i,j) is the first operation executed in machine m:

$$ST_{ij} \geq CT_{hk} - L \times \left(1 - \sum_{m \in M_{ij} \cap M_{hk}} Y_{hkijm}\right) \quad \forall (i,j) \in O, \forall (h,k) \in O \cup \{(0,0)\} \quad (6)$$

Constraint set 7 limits that only one operation can be executed first in each machine:

$$\sum_{(i,j) \in O | m \in M_{ij}} Y_{00ijm} = 1 \qquad \forall m \in M \quad (7)$$

Constraint set 8 guarantees that if the operation (i,j) is performed in machine m then this operation has one predecessor operation in the same machine:

$$X_{ijm} = \sum_{(h,k) \in O \cup \{(0,0)\} | m \in M_{hk}} Y_{hkijm} \qquad \forall (i,j) \in O, \forall m \in M_{ij} \quad (8)$$

Constraint set 9 guarantees that if the operation (i,j) is performed in machine m then this operation has at most one successor operation in the same machine:

$$X_{ijm} \geq \sum_{(h,k) \in O | m \in M_{hk}} Y_{ijhkm} \qquad \forall (i,j) \in O, \forall m \in M_{ij} \quad (9)$$

Constraint set 10 does not allow to put the operation (i,j) after the same or other later processes:

$$\sum_{k \in J_i | k \geq j, m \in M_{ik}} Y_{ikijm} = 0 \qquad \forall (i,j) \in O, \forall m \in M_{ij} \quad (10)$$

Constraint sets 11 to 16 describe the variables' nature:

$$X_{ijm} \in \{0,1\} \qquad \forall (i,j) \in O, \forall m \in M_{ij} \quad (11)$$

$$Y_{hkijm} \in \{0,1\} \qquad \forall (i,j) \in O, \forall (h,k) \in O \cup \{(0,0)\} \forall, m \in M_{ij} \cap \in M_{hk} \quad (12)$$

$$ST_{ij} \geq 0 \qquad \forall (i,j) \in O \quad (13)$$

$$CT_{ij} \geq 0 \qquad \forall (i,j) \in O \cup \{(0,0)\} \quad (14)$$

$$T_i \geq 0 \qquad \forall i \in I \quad (15)$$

$$E_i \geq 0 \qquad \forall i \in I \quad (16)$$

3.1.5. Objective function

The objective function (Equation 17) seeks to minimise the sum of Earliness and Tardiness (ET) of all the jobs.

$$minimise\ Earliness + Tardiness = \sum_{i \in I} (E_i + T_i) \quad (17)$$

3.2. Stochastic processing times

Considering the stochastic environment, two probability distributions were selected to analyse the effect of the stochasticity in the FJSS: log-normal and exponential, which consider only positive values in their domain.

ξPT is defined as the stochastic processing time, and PT as the deterministic processing time which is used as the expected value for the random variable.

- Exponential: Considering the exponential parameter $\lambda = \left(\frac{1}{PT_{ijm}}\right)$, it is possible to model the processing time of the job i in the j-th operation on machine m as shown in Equation 18:

$$\xi PT_{ijm} \sim Exp\left(\frac{1}{PT_{ijm}}\right) \quad (18)$$

- Log-normal: If a random variable X is distributed as a log-normal distribution, then the LnX is normally distributed with mean μ and standard deviation σ. This distribution is used as a model of the influence on environments with uncontrolled variables (Liefooghe et al., 2012).

Considering that the log-normal parameters μ and σ are related to the natural logarithm of the variable, it is possible to calculate these parameters by knowing the expected value and the coefficient of variation (CV) of the variable. See Equations 19 and 20:

$$\mu_{ijm} = Ln\left(PT_{ijm}\right) - \frac{Ln(CV^2 + 1)}{2} \quad (19)$$

$$\sigma_{ijm} = \sqrt{\frac{Ln(CV^2 + 1)}{2}} \quad (20)$$

Thus, it is possible to model the processing time of the job i for the j-th operation on machine m as shown in Equation 21:

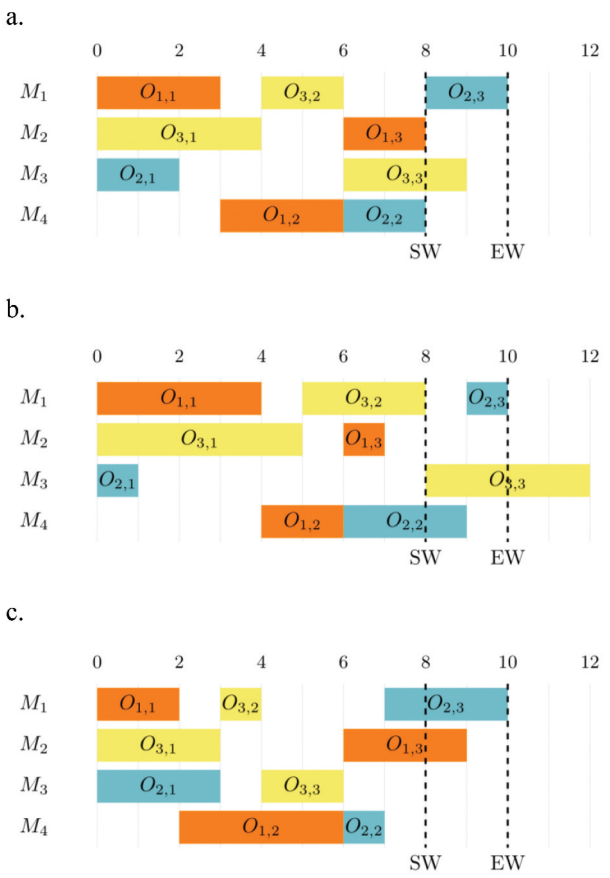


Figure 1. Effect of uncertainty on schedules.

$$\xi PT_{ijm} \sim Log - normal\left(\mu_{ijm}, \sigma_{ijm}\right) \tag{21}$$

Figure 1 shows an example of a schedule with three jobs and four machines, considering the same due window intervals for all jobs. The predictive schedule A has an Earliness+Tardiness of 0 time units, making this sequence an optimal one, from the deterministic point of view. Nevertheless, when the processing times are considered to be stochastic, the schedule performance gets worse obtaining an Earliness+Tardiness of 3 time units and 2 time units for schedules B and C, respectively. In this case, it can be seen that it is very important to take into account the performance of schedules in uncertain environments to improve decision-making.

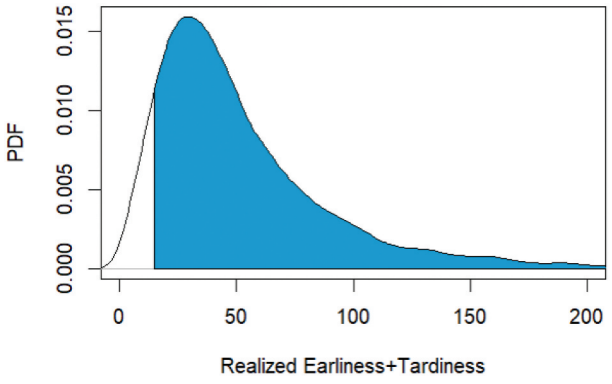


Figure 2. Distribution of realised Earliness+Tardiness.

3.3. Robustness measure

As a result of the stochastic behaviour of the processing times, the completion times of the jobs change, generating a stochastic behaviour in the objective function. Figure 2 shows the distribution of the realised Earliness+Tardiness obtained with a particular scheduling considering exponential processing times. The blue area represents the cases in which the value of the objective function obtains a worse performance than the one calculated with the expected value of the processing times, increasing the related costs. Meanwhile, the white area represents an improvement in this value. Considering this behaviour of the objective function, under random processing times, the measure used by Wu et al. (2020) is adapted for this study.

This Earliness+Tardiness Risk is the average deterioration of the Earliness+Tardiness of the realised schedule (ET^R) and considers stochastic processing times, in comparison to the Earliness+Tardiness of the predictive schedule (ET^P) which uses deterministic processing times. Mathematically, the Earliness+Tardiness Risk (R_{ET}) can be defined as in Equation 22:

$$R_{ET} = E\left[\max\left(ET^R - ET^P, 0\right)\right] \tag{22}$$

Considering the main objective of minimising Earliness+Tardiness, the measure considers as deterioration the cases in which the ET^R is highest than the ET^P, while in the other cases the deterioration is 0 considering that this cases benefits the main objective. As a result, the idea of this robustness measure is to minimise the average deterioration of the schedules under random processing times.

4. The proposed simheuristic: NSGA-II with Monte Carlo simulation

This section describes the proposed simheuristic. Bearing in mind the main elements of the NSGA-II (Deb et al., 2002), the principles of simheuristics (Juan et al., 2015), and the characteristics of the problem; the following aspects are hereafter explained: chromosome definition, initial population, order and selection, crossover operator, mutation operator and Monte Carlo simulation.

Table 3 shows an example of a FJSS problem with a due window interval that is used for explaining the proposed NSGA-II in this section.

4.1. Chromosome definition

Based on L. Sun et al. (2019), the coding used for this problem is a real number from [0, M_{ij}) for

Table 3. Test problem.

Job	Operation	Machines			Due window interval	
		M_1	M_2	M_3	Lower limit	Upper limit
J_1	$O_{1,1}$	2	4	3	10	14
	$O_{1,2}$	3	-	-		
	$O_{1,3}$	3	2	3		
J_2	$O_{2,1}$	3	-	4	12	15
	$O_{2,2}$	-	4	-		
	$O_{2,3}$	3	2	3		
J_3	$O_{3,1}$	2	-	1	8	11
	$O_{3,2}$	4	3	4		

	O_1	O_2	O_3
J_1	2.88	0.19	1.83
J_2	0.28	0.22	1.43
J_3	0.80	2.44	

Figure 3. Representation of the chromosome.

each job operation and each job, M_{ij} being the number of total machines that can process the j-th operation of job i (O_{ij}). The integer part of the number represents the machine selected to process the job. For example, if machines 3, 5, and 7 can process job 2 and the integer part of the gene is 1 (considering 0 as the first position), then, machine 5 is selected, because it is the second machine that can process job 2. The fractional part of the number represents the operations' priority, which is analysed at each step for those operations that can be performed.

Figure 3 shows an example of the representation of the chromosome. The rows correspond to jobs and the columns to the number of the operation. For instance, the cell located in row J_3 column O_1 corresponds to the operation $O_{3,1}$. Therefore, the allocation sequence represented in this example is as follows:

- $O_{1,1}$ is processed in its third machine, considering that its fractional part is the greatest among all the operations that can be processed ($O_{1,1}$: 0.88, $O_{2,1}$: 0.28 and $O_{3,1}$: 0.80).
- $O_{3,1}$ is processed in its first machine, considering that its fractional part is the greatest among all the operations that can be processed ($O_{1,2}$: 0.19, $O_{2,1}$: 0.28 and $O_{3,1}$: 0.80).
- $O_{3,2}$ is processed in its third machine, considering that its fractional part is the greatest among all the operations that can be processed ($O_{1,2}$: 0.19, $O_{2,1}$: 0.28 and $O_{3,2}$: 0.44).
- $O_{2,1}$ is processed in its first machine, considering that its fractional part is the greatest among all the operations that can be processed ($O_{1,2}$: 0.19 and $O_{2,1}$: 0.28).
- $O_{2,2}$ is processed in its first machine, considering that its fractional part is the greatest among all the operations that can be processed ($O_{1,2}$: 0.19 and $O_{2,2}$: 0.22).
- $O_{2,3}$ is processed in its second machine, considering that its fractional part is the greatest among all the operations that can be processed ($O_{1,2}$: 0.19 and $O_{2,3}$: 0.43).
- $O_{1,2}$ is processed in its first machine, considering that its fractional part is the greatest among all the operations that can be processed (only $O_{1,2}$: 0.19).
- The last remaining operation $O_{1,3}$ is processed in its second machine.

Figure 4 shows the Gantt chart of the example which obtains a Earliness+Tardiness of 4 time units.

Finally, L. Sun et al. (2019) considers two important points in the design of the representation of a solution. The first one is that each representation has to be as short as possible, thus reducing the different variables to evaluate. As a second point, each representation has to be a feasible solution to guarantee the efficiency of the algorithm, reducing the computational execution time by omitting the unfeasible operations. In this case, with the proposed coding, if the integer part is changed the machine on which the operation is processed also changes, taking into account that the range only considers the machines that can process the j-th operation of job i. Moreover, if the fractional part

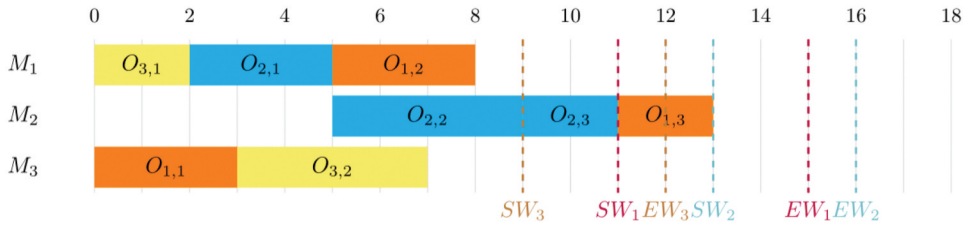


Figure 4. Gantt chart of the chromosome solution of Figure 3 using the instance in Table 3.

is modified, the priority against the other operations also varies.

4.2. Initial population

For the initial population of the proposed NSGA-II, some solutions were generated with the application of three static dispatching rules and three dynamic dispatching rules considering the expected values of the processing times. The remaining solutions to reach the number of individuals were randomly generated.

At each iteration, all jobs consider processing their current operation in the machine with the lowest availability time. If there are two or more machines with this characteristic, the machine with the lowest processing time is selected. If there is a tie between two or more machines, one of them is chosen randomly. Once all jobs have a candidate machine to process them, the selection criterion to assign a job is, on the one hand, the job with the lowest availability time (considering the availability time of the machine when it finishes the previous operation). On the other hand, if there is a tie between two or more jobs, the following dispatching rules are used to break the tie:

- EDD (earliest due date): Select job i if it has the minimum value of EW_i
- SPT (shortest processing time): Select job i if it has the minimum value of PT_{ijm} for the current operation j in the selected machine m.
- LPT (longest processing time): Select job i if it has the maximum value of PT_{ijm} for the current operation j in the selected machine m.
- CR (critical ratio): Select job i if it has the minimum value of CR_i for the current operation j, as in Equation 23:

$$CR_i = max\left(\frac{SW_i - ST_{ij}}{\sum_{h=j}^{J_i}\overline{PT_{ij*}}}, 0\right) \tag{23}$$

where $\overline{PT_{ij*}}$ is the processing times average of all the possible machines that can perform the j-th task.

- AVPRO(average processing time per operation): Select job i if it has the minimum value of AP_i for the current operation j, as in Equation 24:

$$AP_i = \frac{\sum_{h=j}^{J_i}\overline{PT_{ih*}}}{|J_i| - j + 1} \tag{24}$$

where $\overline{PT_{ih*}}$ is the processing times average of all the possible machines that can perform the h-th task.
- S/RO(slack per remaining operations): Select job i if it has the minimum value of SR_i for the current operation j, as in Equation 25:

$$SR_i = \frac{max\left(SW_i - CT_{i(j-1)}, 0\right)}{|J_i| - j + 1} \tag{25}$$

Finally, if there remains a tie between jobs, a job is selected randomly to schedule it in the selected machine and update the current operation of the job and the availability times. This process has to be done until all the operations have been assigned.

4.3. Order and selection

After the initial population is generated, there is a group of N chromosomes that are chosen to be recombined. This group will be classified into Pareto frontiers, according to the fast non-dominated sorting procedure. In this procedure, each solution is compared with every other solution in the population to find those that are non-dominated. Once a set of non-dominated solutions is found, these solutions are assigned to a first front (F_1). Subsequently, the remaining solutions that were not classified in the F_1 front are compared with each other once again to select those that are non-dominated, to classify them into the F_2 front. This procedure is repeated successively until all the F_n fronts are found, i.e., all the chromosomes are classified entirely. Later, in each Pareto frontier, the chromosomes are ordered according to their crowding distance, calculated according to Algorithm 1 for each set I.

Algorithm 1

Crowding_distance_assignment(I) :

```
s = |I|                                   ▷ Number of solutions in the Pareto Frontier.
for each I, set Distance[i] = 0           ▷ Initialise distance.
for each objective function m do
  I = sort(I, m)                          ▷ Sort I according each objective function.
  Distance[1] = Distance[s] = ∞           ▷ Extreme points are always selected.
    for i = 2 to (s − 1) do               ▷ For the other points.
      Distance[i] = Distance[i] + I[i + 1].m − I[i − 1].m
    end for
end for
```

	O_1	O_2	O_3
J_1	2.88	0.19	1.83
J_2	0.28	0.22	1.43
J_3	0.80	2.44	

	O_1	O_2	O_3
J_1	0.27	0.44	2.47
J_2	1.38	1.33	2.96
J_3	0.33	2.21	

	O_1	O_2	O_3
J_1	2.88	0.19	2.47
J_2	1.38	0.22	1.43
J_3	0.33	2.44	

Figure 5. Crossover operation.

Furthermore, the crowding distance assigned to the solutions at the extremes of the frontier is a considerable number (∞). Higher crowding distances are given more weight because the farther apart the chromosomes are, the more diversification occurs during crossover.

Once all the chromosomes are ordered, a pair of chromosomes is selected randomly to apply the crossover operator until Q offspring is obtained. Next, a population with size N+Q is obtained. Later, the chromosomes of the population N+Q are organised again by the fast-non-dominated sorting and crowding distance procedures, and the first N chromosomes are saved. It is important to emphasise that the previous process is repeated until the stopping criterion is met.

4.4. Crossover operator

A random number between 0 and 1 is generated for each position of the chromosome, which defines how the crossover will be performed to create an offspring chromosome. When the random number is less than the crossover probability, the offspring takes the value of the first parent's gene in this position. Otherwise, it takes the value of the second parent's gene in this position.

In the example of Figure 5, the yellow forms represent the parent chromosomes, and the red form represents the offspring chromosome. Figure 6 shows the Gantt chart of the example, which obtains a Earliness+Tardiness of 1 time unit.

4.5. Mutation operator

Initially, a probability of mutation is defined. For each resulting chromosome, a random number between 0 and 1 is generated to determine whether the mutation operator should be applied to that chromosome or not. If the generated number is less than the defined probability, the mutation is performed. The mutation consists of choosing one job of the chromosome and reversing the order of the fractional values of the operations. In the example of Figure 7, the obtained offspring inverts the order of the fractional values of Job 3, from 0.33 and 2.44 to 0.44 and 2.33, respectively.

Figure 8 shows the Gantt chart of the example which obtains a Earliness+Tardiness of 3 time units.

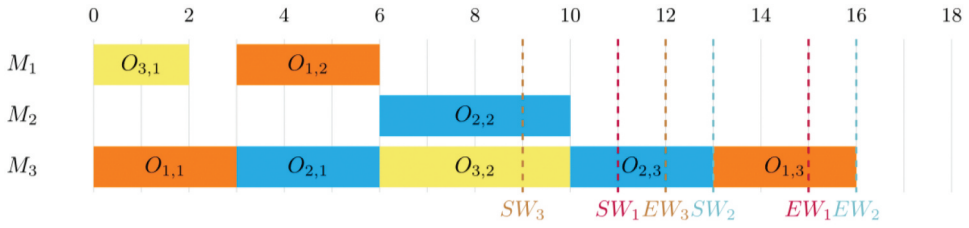


Figure 6. Gantt chart of the chromosome solution of Figure 5 using the instance in Table 3.

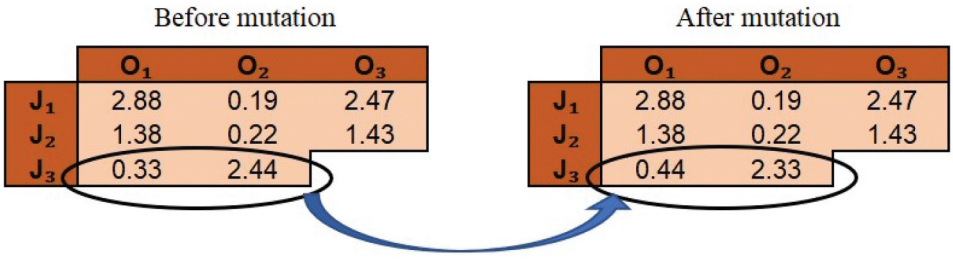


Figure 7. Perform of the mutation.

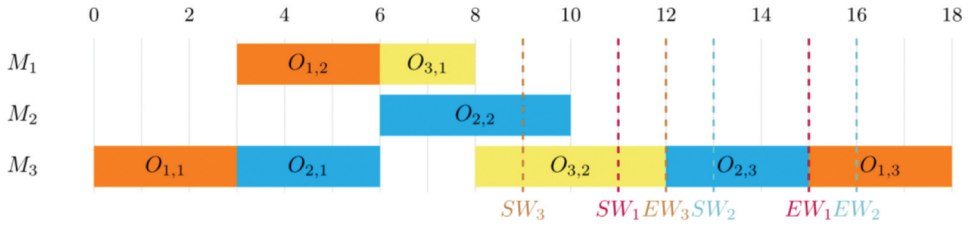


Figure 8. Gantt chart of the chromosome solution of Figure 7 using the instance in Table 3.

4.6. Monte Carlo simulation

According to Juan et al. (2015), a simheuristic is an algorithm that integrates any simulation variant into a metaheuristic framework to solve complex stochastic optimisation problems. Thus, each solution generated through NSGA-II is simulated through a Monte Carlo simulation to evaluate the stochastic objective function, using the probability distributions defined for the stochastic parameters.

The author indicates that only promising deterministic solutions should be simulated in the simheuristic process, in order to reduce computational times. However, this statement has been modified for this study considering the requirement of having the deterministic and stochastic objective functions of each solution to perform NSGA-II to avoid the risk of losing any solution that may be suitable for the Pareto frontier.

Thus, following the proposed methodology, during the iterative search process the number of replications of the simulations should be small (fast simulation) to obtain approximate estimates reducing the computational time. Besides, at each generation, only the new chromosomes in the population are simulated with fast simulation because those that are not new (i.e., that came from previous generations) were simulated when they appeared for the first time in the

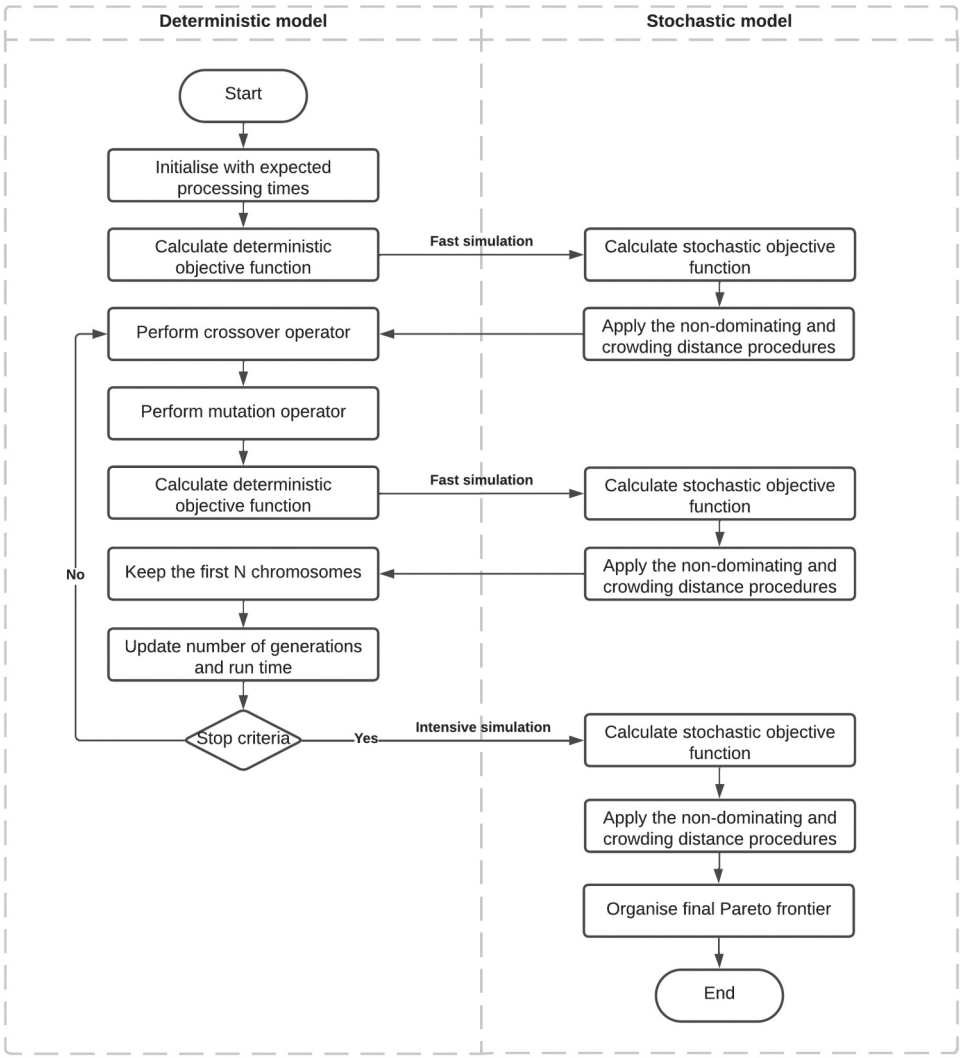


Figure 9. Flow chart diagram of the simheuristic algorithm.

population. Once the NSGA-II procedure meets the stopping criterion, all the chromosomes obtained in the last generation have to be simulated again, but, in this case, the number of simulation replicates has to be extensive (intensive simulation) to obtain high-quality estimates and to reorder all the solutions for the final Pareto frontier.

4.7. Simheuristic flow chart

The proposed simheuristic follows the procedure shown in Figure 9, which is based on the methodology proposed by Juan et al. (2015):

5. Computational experiments

This section presents the computational experiments carried out to evaluate the performance of the proposed simheuristic. Since there are not similar works with the same objective functions for comparison, this section reports various computational experiments. First, the performance of the simheuristic is evaluated by comparing the simulations given by three different deterministic solution approaches to the problem: the MILP model, dispatching rules and single-objective genetic algorithm. Secondly, the visualisation of some of the obtained Pareto frontiers is shown. Thirdly, the behaviour of the objective functions is analysed under different coefficients of variation and probability distributions. Finally, an experiment is carried out to show the effects of different probability distributions and coefficients of variation in multi-objective measures of Pareto frontiers.

None of the experimental designs performed in this section fulfilled the homoscedasticity and normality assumptions of ANOVA. Therefore, the non-parametric test MANOVA (Anderson, 2001) was executed in each case, and all conclusions are presented with a significance level of 0.05.

5.1. Parameters settings

The simheuristic algorithm is coded and executed by using NetBeans IDE 8.2 on an Intel(R) Core™ i7 CPU@2.20 GHz with 7.84 GB RAM. The parameter values chosen for the NSGA-II (number of chromosomes, number of offspring, number of generations, crossover probability and mutation probability) were selected through an experimental design, and the rest were selected based on experience and some references from the reviewed literature. Table 4 shows the values selected for the parameters.

Table 4. Simheuristic parametrisation.

Parameter	Value
Number of chromosomes	1000
Number of offspring	500
Number of generations	750
Crossover probability	30%
Mutation probability	2.5%
Replicates (Fast simulation)	50
Replicates (Intensive simulation)	1000
Maximum time of execution (minutes)	60

5.2. Benchmark instances

Because of the absence of standard benchmark instances for different variations of the FJSS under stochastic factors, some deterministic FJSS instances were selected and adapted considering the deterministic processing times as the expected values of the random processing times. Two sets of benchmarks were evaluated to analyse the performance of the proposed simheuristic:

- Brandimarte (1993), p. 10 instances (Mk01 to Mk10) with sizes ranging from 10 jobs and 6 machines to 20 jobs and 15 machines.
- Hurink et al. (1994), p. 56 instances (abz5 to abz9, car1 to car8, la01 to la40, mt06, mt10 and mt20) with sizes ranging from 6 jobs and 6 machines to 30 jobs and 10 machines.

Considering that these instances do not have due windows, their lower limits were generated using the method described by Bagheri and Zandieh (2011) as shown in Equation 26.

$$SW_i \sim U\left[\mu_i \times \left(1-\frac{R}{2}\right); \mu_i \times \left(1+\frac{R}{2}\right)\right] \tag{26}$$

where μ_i is calculated as shown in Equation 27,

$$\mu_i = \left(1+\frac{T\times I}{M}\right) \times \sum_{j=1}^{J_i} \overline{PT_{ij*}} \tag{27}$$

Considering I as the number of jobs, M the number of machines, and T and R as parameters related with the complexity of the due dates, the same values used by Bagheri and Zandieh (2011) were used. $\overline{PT_{ij*}}$ is the processing times average of all the possible machines that can perform the j-th task.

The upper limit of due windows was calculated according to Equation 28:

$$EW_i \sim U[SW_i + 0.1 \times \beta; SW_i + \beta] \tag{28}$$

where β is calculated as half of the minimum mean total processing time per operation, as shown in Equation 29.

$$\beta = \frac{min\left(\sum_{j=1}^{J_i} \overline{PT_{ij*}}\right)}{2} \tag{29}$$

5.3. Mixed integer linear programming

The proposed MILP model was executed in Jupyter notebook with Python 3 and using the PuLP library,

Table 5. Results of MILP model.

		Earliness+Tardiness				Earliness+Tardiness Risk		
Instance	State	MILP	SH	Improv	PD	MILP	SH	Improv
car7	F	11446	0	1.00%	LogN25	1703.89	22.77	98.66%
car7	F	11446	0	1.00%	LogN50	5376.54	117.13	97.82%
car7	F	11446	0	1.00%	LogN75	10527.93	794.44	92.45%
car7	F	11446	0	1.00%	LogN100	16627.19	2448.48	85.27%
car7	F	11446	0	1.00%	Exp	12221.26	1817.58	85.13%
la04	F	2973	243	91.83%	LogN25	217.35	46.00	78.84%
la04	F	2973	250	91.59%	LogN50	664.01	178.18	73.17%
la04	F	2973	237	92.03%	LogN75	1289.82	401.03	68.91%
la04	F	2973	223	92.50%	LogN100	2051.75	590.26	71.23%
la04	F	2973	243	91.83%	Exp	1372.13	548.65	6.01%
Mk01	F	702	0	1.00%	LogN25	25.45	0.09	99.66%
Mk01	F	702	1	99.86%	LogN50	72.43	2.64	96.35%
Mk01	F	702	0	1.00%	LogN75	143.58	13.82	9.37%
Mk01	F	702	0	1.00%	LogN100	226.21	26.82	88.14%
Mk01	F	702	1	99.86%	Exp	107.94	24.27	77.52%
mt06	Opt	0	0	.00%	LogN25	5.98	0.24	95.91%
mt06	Opt	0	0	.00%	LogN50	21.69	2.24	89.66%
mt06	Opt	0	0	.00%	LogN75	5.24	10.59	78.92%
mt06	Opt	0	0	.00%	LogN100	82.97	17.79	78.56%
mt06	Opt	0	0	.00%	Exp	79.11	16.75	78.82%

in the computer mentioned in subsection 5.1. All the benchmark instances were tested with a time limit of 8 h. Considering this time constraint, three states can be obtained for each instance": non-feasible", which shows that the model was not able to obtain a solution satisfying all constraints in the time frame mentioned above"; feasible", which indicates that a solution that satisfies all constraints was obtained, but it is not guaranteed that this best possible value for the objective function; and "optimal", which states that a solution with the best possible value for the objective function was found.

The instances that obtained a feasible (F) or optimal (Opt) solution with the MILP model were simulated using the log-normal distribution (LogN), with coefficients of variation of 25%, 50%, 75%, and 100%, and the exponential distribution (Exp), to obtain the Earliness+Tardiness Risk in each case. The results were compared with the solutions obtained with the proposed simheuristic (SH) for the same distributions and coefficients of variation. Table 5 shows the results for all instances that obtained a feasible or optimal solution with the MILP model for each probability distribution (PD).

The simheuristic achieves the same Earliness +Tardiness as the simulated solution given by the MILP model for the instance that was optimal, and improved by 97.30% the Earliness+Tardiness of the instances that obtained a feasible solution. Finally, the simheuristic improves by an average of 84.27% the Earliness+Tardiness Risk in comparison with the results obtained with the simulation of the solutions given by the MILP.

5.4. Dispatching rules

From the previous results, it is concluded that trying to use the proposed MILP model to obtain the optimal or some feasible solutions requires unreasonable computational times in most instances. In this case, the use of dispatching rules ensures feasible solutions that can be obtained in less time. An experimental design was carried out to measure the performance of the simheuristic in comparison with simulated solutions of six dispatching rules considering the following factors: the combination of the probability distribution with the coefficient of variation (Log-normal25, Log-normal50, Log-normal75, Log-normal100 and Exponential), dispatching rules (EDD, SPT, LPT, CR, AVPRO and S/RO) and instances as a blocking factor. The response variable was the improvement in the value of each objective function m, calculated as the percentage difference between the best value obtained for that objective function with the simheuristic and the value obtained for the same objective function with the dispatching rule, as shown in Equation 30.

$$Improvement_m = \frac{Dispatch\ Rule_m - Best_m}{Dispatch\ Rule_m} \in Objective\ Functions \tag{30}$$

Table 6. Non-parametric MANOVA for improvement in each objective function.

Factor	ET	R_{ET}
Instance	0.0000	0.0000
Combination	0.0000	0.0000
Dispatching rule	0.0000	0.0000
Instance*Combination	0.0000	0.0000
Instance*Dispatching rule	0.0000	0.0000
Combination*Dispatching rule	0.9321	0.0000
Instance*Combination*Dispatching rule	1.0000	0.0000
R^2_{aj}	95.69%	97.88%

Table 6 shows the p-values and the coefficients of determination obtained for the improvements in each

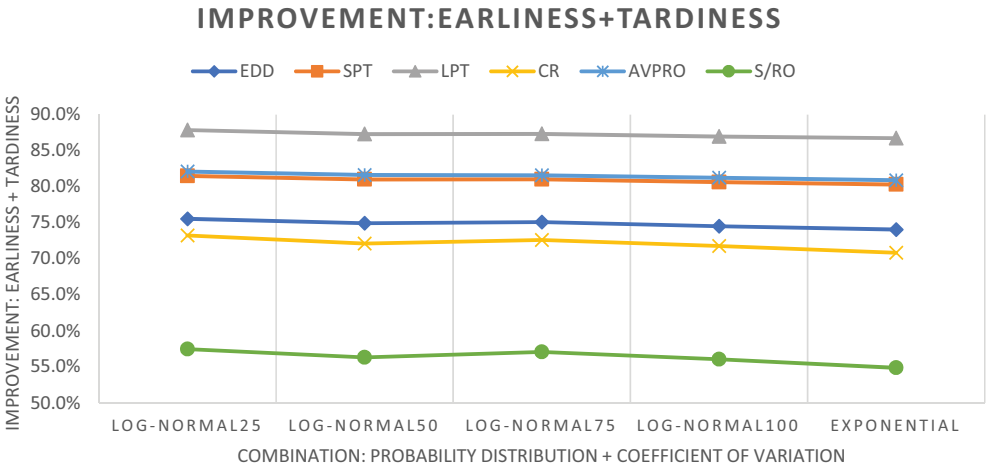


Figure 10. Profile chart of Earliness+Tardiness improvement.

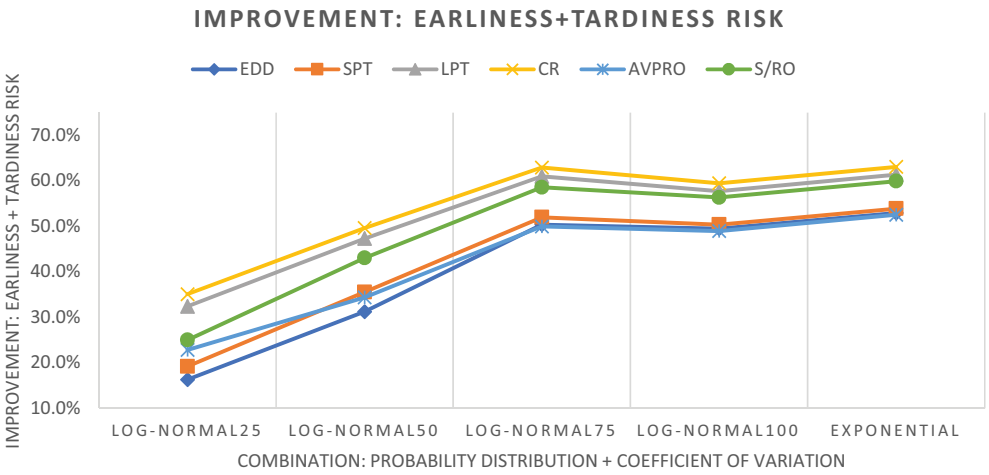


Figure 11. Profile chart of Earliness+Tardiness Risk improvement.

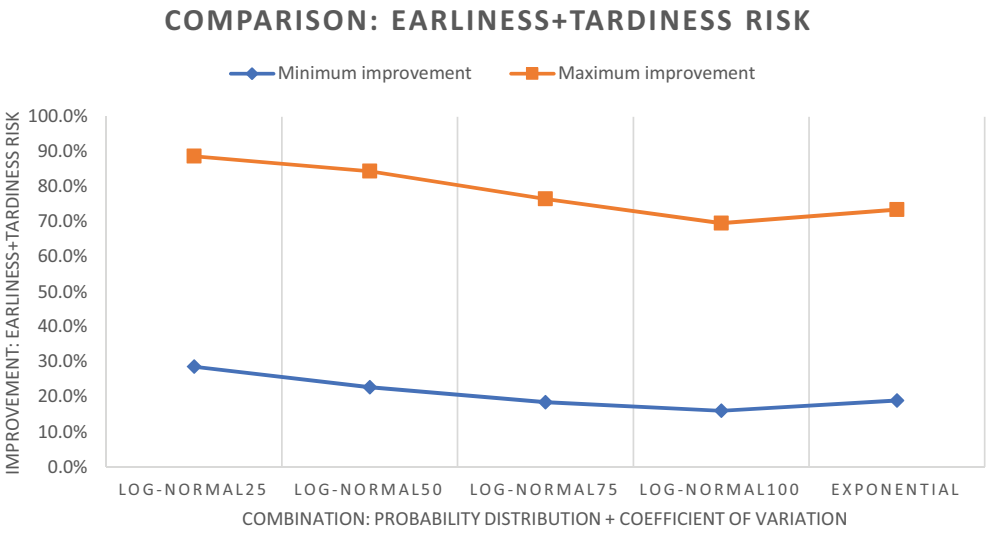


Figure 12. Profile chart of Earliness+Tardiness Risk improvement against genetic algorithm.

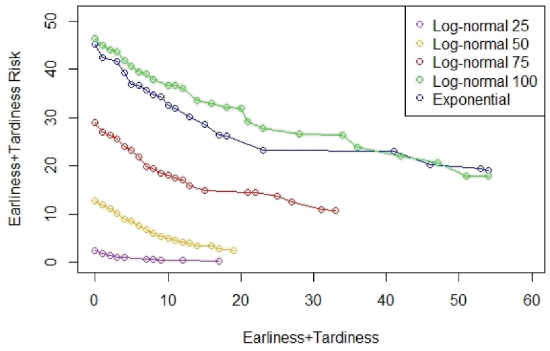
objective function with the non-parametric MANOVA.

According to MANOVA, the first factor (which is the combination of the coefficient of variation and the probability distribution) and the second factor (which is the dispatching rule) have both a significant effect in the improvement measure of both objective functions. However, the interaction of those two factors has a significant effect only in the improvement measure of the stochastic objective function.

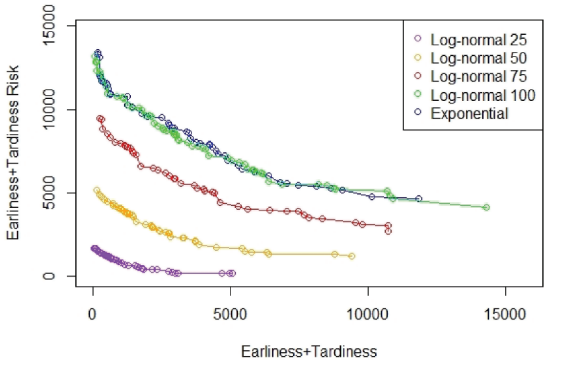
Average improvements can be observed in Figures 10 and 11 for Earliness+Tardiness and Earliness+Tardiness Risk, respectively. It can be seen that for the Earliness+Tardiness objective the best improvement is reached for lower coefficients of variation whereas for Earliness+Tardiness Risk the best improvements are obtained for higher coefficients of variation. In the case of dispatching rules, the least improvement of Earliness+Tardiness is about 50% against the SR/O dispatching rule and for Earliness+Tardiness Risk is a minimum of 15% against the EDD dispatching rule.

5.5. Deterministic approach

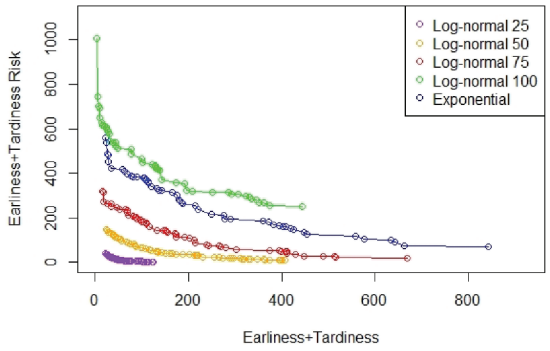
Another approach that is more efficient in terms of computational time is to solve the deterministic version of the problem using the single-objective genetic algorithm and then simulating the best solution obtained to calculate the robustness measure. Using the same



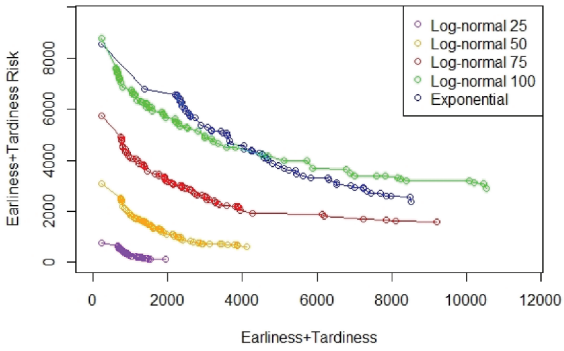
(a) Pareto frontiers obtained for mt06 instance.



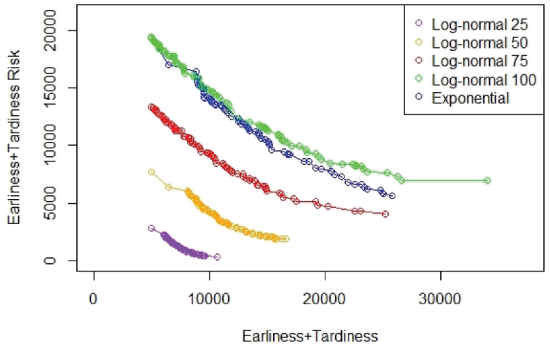
(b) Pareto frontiers obtained for car5 instance.



(c) Pareto frontiers obtained for Mk03 instance.



(d) Pareto frontiers obtained for la30 instance.



(e) Pareto frontiers obtained for la34 instance.

Figure 13. Pareto frontiers obtained.

parameters as above, a genetic algorithm is executed, and, at the end, the best solution is simulated with a total of 1000 replicates considering a combination of the probability distribution with the coefficient of variation (Log-normal25, Log-normal50, Log-normal75, Log-normal100 and Exponential). Then, these results are compared with the best Earliness+Tardiness Risk obtained by the simheuristic approach, calculating the improvement as in Equation 31.

$$Improvement_{R_{ET}} = \frac{GeneticAlgorithm_{R_{ET}} - Best_{R_{ET}}}{GeneticAlgorithm_{R_{ET}}} \quad (31)$$

In the main effects plot of Earliness+Tardiness Risk improvement (Figure 12), it can be observed that the solution with the best and the worst Earliness+Tardiness Risk given by the simheuristic approach improves the Earliness+Tardiness Risk provided by the metaheuristic approach in a range from 69.62% to 88.73% and from 16.03% to 28.65%, respectively, according to the combination of the probability distribution and coefficient of variation. This shows that the GA is not able to deal with the impact of variability of processing times in the Earliness+Tardiness, whereas, at the moment of including an objective function that evaluates the robustness, the simheuristic obtains solutions that minimise the impact of uncertainties in the performance of Earliness +Tardiness.

5.6. Pareto frontiers

A total of 1650 Pareto frontiers were obtained. They consist of five frontiers for each instance, one each for the combination of the probability distribution with the coefficients of variation (Log-normal25, Log-normal50, Log-normal75, Log-normal100 and Exponential) of the processing times.

As an example, the five Pareto frontiers for the instances mt06 (Figure 13a), car5 (Figure 13b), Mk03 (Figure 13c), la30 (Figure 13d) and la34 (Figure 13e) are presented. These instances consist of 6, 10, 15, 20 and 30 jobs, respectively. In these examples, it can be seen that there is a contradictory behaviour of the two objectives evaluated, and that the minimum value obtained for the Earliness+Tardiness Risk increases as the coefficient of variation increases.

This behaviour can be observed in the Pareto frontiers of the other 61 instances. Nevertheless, it is important to perform a statistical analysis of the results and the behaviour of the Pareto frontier. The experiments carried out for the results obtained are explained in the following subsections.

Table 7. Non-parametric MANOVA for the objective functions with different coefficients of variation.

Factor	ET	R_{ET}
Instance	0.0000	0.0000
CV	0.7514	0.0000
Instance*CV	0.0000	0.0000
R^2_{aj}	99.49%	98.68%

Table 8. Non-parametric MANOVA for the objective functions with different probability distributions.

Factor	ET	R_{ET}
Instance	0.0000	0.0000
Distribution	0.4466	0.0000
Instance*Distribution	0.7953	0.0000
R^2_{aj}	99.56%	98.32%

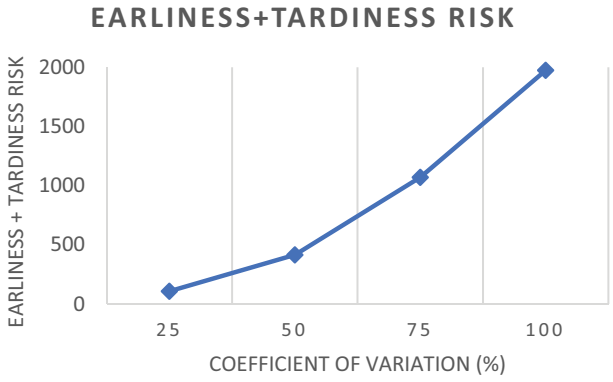


Figure 14. Profile chart of Earliness+Tardiness Risk.

5.7. Objective functions

It is also essential to analyse the behaviour of objective functions under different probability distributions and coefficients of variation. Considering that the log-normal distribution was tested with coefficients of variation of 25%, 50%, 75%, and 100% and that the exponential distribution has a single coefficient of variation of 100%, two analyses were performed: i) comparison of different coefficients of variation under a log-normal distribution and ii) comparison of a log-normal distribution with a coefficient of variation of 100% and exponential distribution.

Table 7 shows the p-values and coefficients of determination obtained for each objective function, considering different coefficients of variation (CV) under a log-normal distribution with the non-parametric MANOVA.

According to the results obtained, the coefficient of variation has a significant effect on the stochastic objective function. According to the main effects plots and least significant difference test (LSD), with a significance of 95%, it can be observed that the Earliness+Tardines Risk increases as the coefficient of variation augments (Figure 14).

Table 8 shows the p-values and the coefficients of determination obtained with the MANOVA for both objective functions under different probability distributions (with CV = 100%).

According to the results obtained, the probability distribution (under a coefficient of variation of 100%) has a significant effect on the value of Earliness +Tardiness Risk. Specifically, the exponential distribution presents a better average of Earliness+Tardiness Risk (1766.96) than the log-normal distribution (1974.254), even though the coefficient of variation is the same.

These tests allow us to highlight the relevance of performing stochastic analysis (using of simheuristics) to deal with uncertainties and, on the other hand, the importance of a precise adjustment of the probability distribution used in the method in order to obtain good and accurate results.

5.8. Performance measures

Finally, it is important to evaluate the performance of the NSGA-II algorithm in terms of the quality of the obtained Pareto frontiers. For this purpose, the performance criteria proposed by Ahmadi et al. (2016) were evaluated:

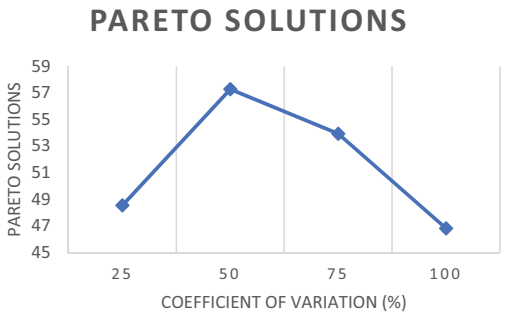
- Number of Pareto solutions (NP): the number of solutions obtained on the first Pareto frontier of the last generation of the algorithm. High values of the obtained solutions imply more varied options for the decision-making process.
- Spacing: the standard deviation of the nearest absolute distance from one solution to another (see Equation 32). This determines the consistency between the Pareto solutions, which means that the lower this variability between the distances of the Pareto frontier solutions is, the higher the quality of the algorithm results.

$$Spacing = \sqrt{\frac{1}{NP-1}\sum_{i=1}^{NP}\left(d_i - \overline{d}\right)^2} \qquad (32)$$

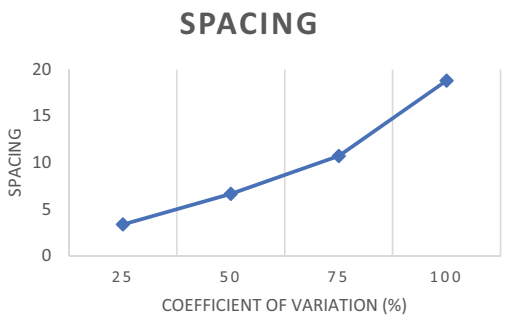
where d_i is the minimum absolute distance between the solution i and another solution, and $\overline{d}$

Table 9. Non-parametric MANOVA for performance measures with different coefficients of variation.

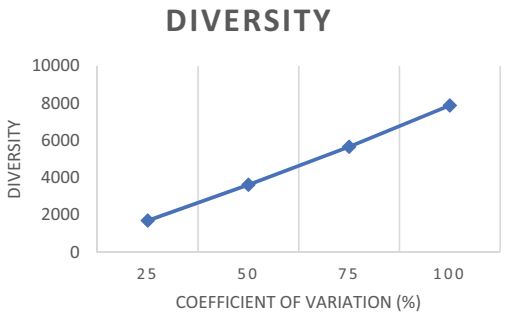
Factor	NP	Spacing	Diversity	MMID	Run time
Instance	.0000	.0000	0.0000	0.0000	0.0000
CV	.0000	.0000	0.0000	0.0000	0.3725
Instance*C.V.	.0000	.0000	0.0000	0.0000	1.0000
R^2_{aj}	75.61%	74.66%	92.16%	51.61%	98.35%



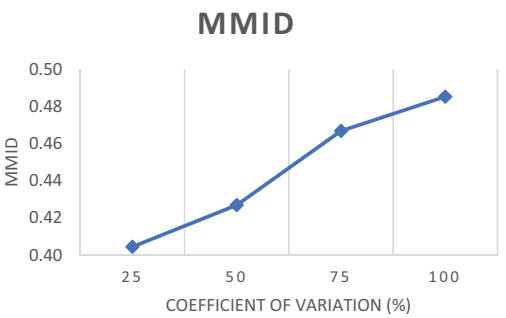
(a) Profile chart of Pareto solutions.



(b) Profile chart of spacing.



(c) Profile chart of diversity.



(d) Profile chart of MMID.

Figure 15. Profile charts.

Table 10. Non-parametric MANOVA for performance measures with different distributions.

Factor	NP	Spacing	Diversity	MMID	Run time
Instance	.0000	.0000	0.0000	0.0000	0.0000
Distribution	.0000	.0000	0.0076	0.0008	0.0000
Instance*Distribution	.0841	.0520	0.0000	0.0659	0.0000
R^2_{aj}	74.27%	7.47%	91.03%	46.57%	99.32%

Table 11. Performance measures by probability distribution.

	Distribution	
Measure	Log-normal	Exponential
Pareto solutions	46.812	56.458
Spacing	18.810	13.322
Diversity	7882.481	7355.057
MMID	0.485	.473
Run time (min)	10.555	7.790

is the average of all the minimum distances of all the solutions.

- Diversity: the Euclidean distance between the most extreme values of the obtained Pareto frontier (see Equation 33).

$$Diversity = \sqrt{(MaxET - MinET)^2 + (MaxR_{ET} - MinR_{ET})^2} \quad (33)$$

A higher value of the diversity of the frontier represents a higher quality in the set of solutions found.

- Modified mean ideal distance (MMID): the average distance of the normalised points, considering the difference of scales in the objective functions, to an ideal point $(0, 0)$. The MMID is calculated as shown in Equation 34:

$$MMID = \frac{1}{NP}\sum_{i=1}^{NP}\sqrt{NET_i^2 + NR_{ET}{}_i^2} \quad (34)$$

where NET and NR_{ET} are calculated, as presented in Equations 35 and 36, taking into account the minimum and maximum values of each objective function:

$$NET_i = \frac{ET_i - MinET}{MaxET - MinET} \quad (35)$$

$$NR_{ETi} = \frac{R_{ETi} - MinR_{ET}}{MaxR_{ET} - MinR_{ET}} \quad (36)$$

This criterion evaluates the efficiency and convergence power of the algorithm and considering that both objectives are to be minimised, lower values indicate a higher quality algorithm performance.

- Run time: the execution time of the algorithm. Obtaining a good response in a short time represents the high quality of the algorithm to cope with emergency situations.

Table 9 shows the p-values and the coefficients of determination obtained for each performance measure with different coefficients of variation obtained with the non-parametric MANOVA.

As shown by the results, the coefficient of variation has a significant effect on all the performance measures. In the main effects plots of the number of Pareto solutions (Figure 15a), spacing (Figure 15b), diversity (Figure 15c) and MMID (Figure 15d), and employing the LSD test with a significance of 95%, it can be observed that the spacing, diversity and MMID increase as the coefficient of variation increases. For the Pareto solutions measure, the best value is reached with a coefficient of variation of 50%, followed by a coefficient of variation of 75% and, finally, the lowest value is obtained with coefficients of variation of 25% and 100%.

Table 10 shows the p-values and the coefficients of determination obtained for each performance measure with different probability distributions (log-normal and exponential) under the same coefficient of variation (100%) with the non-parametric MANOVA.

According to the results obtained, the type of probability distribution has a significant effect on all the performance measures. In Table 11, it is observed that the exponential distribution performs better in all measures except diversity compared to the log-normal distribution.

6. Conclusions and future work

This research project addresses a bi-objective flexible job shop problem that seeks to minimise the earliness and tardiness in the deterministic environment and to minimise the risk associated with the randomness of processing times in a stochastic environment to obtain robust solutions. For this purpose, a simheuristic was developed that hybridises the multi-objective genetic algorithm NSGA-II and Monte Carlo simulation to obtain a Pareto frontier of the two evaluated objective functions. The performance of the proposed algorithm was evaluated in two stages. In the first stage, the results obtained with the proposed simheuristic were compared with the results obtained by other approaches that provide a particular solution for the problem. Those approaches were as follows: simulation of the solutions obtained with the MILP model, simulation of the solutions achieved with the EDD, SPT, LPT, CR, AVPRO and S/RO dispatching rules and simulation of the solutions obtained with the deterministic single-objective version of the genetic algorithm. The second stage presents the analysis of the influence of the coefficients of variation and probability distributions on the evaluated objective functions and the resulting Pareto frontier performance measures.

The comparison of the simheuristic against the simulation of the solutions obtained with the MILP model shows that the simheuristic performs better than the simulated MILP model in terms of computational cost, obtaining better results, though not necessarily the optimum, in less execution time. Against the simulation of the solutions achieved with the mentioned dispatching rules, the simheuristic improves on average by at least 55% the Earliness+Tardiness and on average by at least 15% the Earliness+Tardiness Risk compared to the evaluated dispatching rules. And against the simulation of the solutions obtained with the deterministic single-objective version of the genetic algorithm, the simheuristic improves by at least 69% the Earliness+Tardiness Risk. Lastly, in the second stage, an influence of the coefficients of variation on the value of both objective functions and most of the performance measures analysed is observed when the probability distribution does not vary. Meanwhile, there is a significant difference between the exponential and log-normal distributions under a coefficient of variation of 100% that is reflected in the stochastic objective function and all the performance measures analysed. The above results demonstrate the importance of including stochasticity in the decision-making process, and of making an accurate distribution fitting in order to obtain consistent and appropriate results.

For future research, it is recommended to collect data from a realistic FJSS, perform probability distribution fitting and evaluate the performance of the simheuristic in comparison to dispatching rules with the same fitted probability distributions. In addition to this, it is suggested to integrate other stochastic factors that may occur in a manufacturing environment, such as machine breakdowns, job reprocessing, and job and machine availability time variation, among others. Another topic that can be studied is the analysis of the behaviour of other probability distributions under similar conditions, such as the gamma or Weibull distributions. An additional exploration is to evaluate this problem with other simheuristic modelling that uses Monte Carlo simulation hybridised with other metaheuristics, such as tabu search, PSO, and iterated local search, among others. We also suggest hybridising the proposed NSGA-II with other types of simulation, such as agent-based or discrete-event, and making comparisons with the proposed simheuristic. Finally, it is recommended to test the computing budget allocation or the implementation of common random numbers to make the simheuristic more efficient.

Disclosure statement

No potential conflict of interest was reported by the author(s).

ORCID

Eliana María González-Neira http://orcid.org/0000-0002-4590-3401

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